

Leishmaniases

Parasite: Leishmania (>20 species)

Hosts: Humans and animals (dogs)

Reservoir hosts: Dogs, rodents, hyraxes, etc...

Biological vector: Sand flies

Transmission: Anthroponosis/zoonoses

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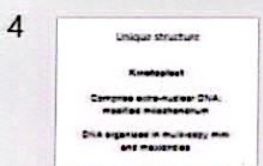
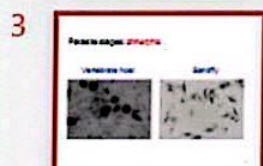
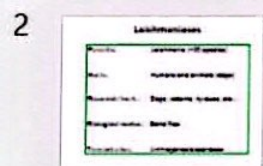
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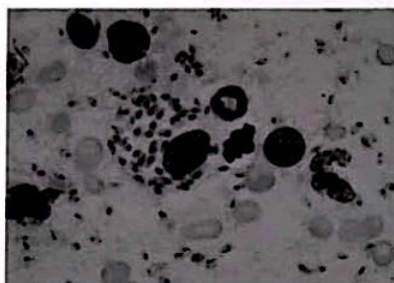
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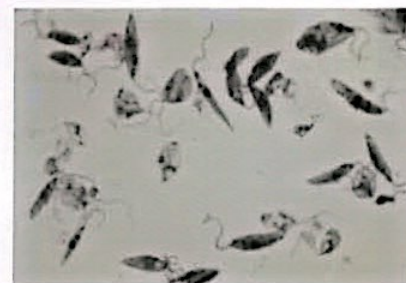


Parasite stages: **dimorphic**

Vertebrate host

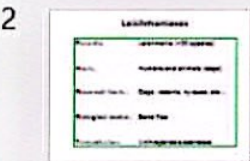
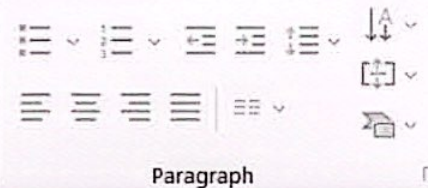
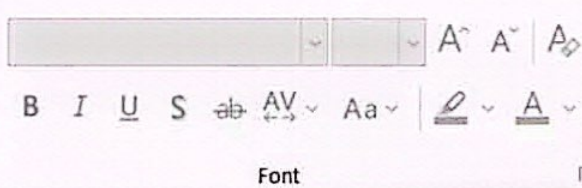
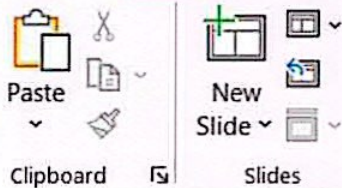


Sandfly



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Unique structure

Kinetoplast

Comprise extra-nuclear DNA;
modified mitochondrion

DNA organized in multi-copy mini
and maxicircles

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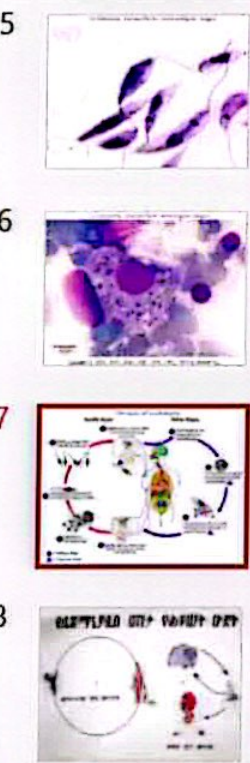
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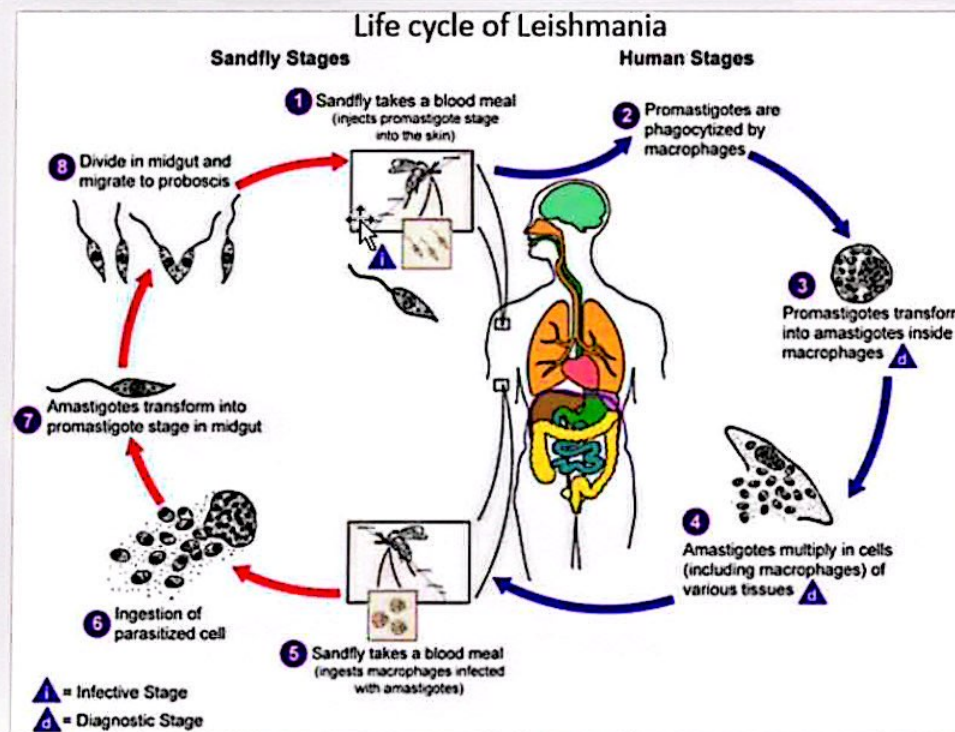
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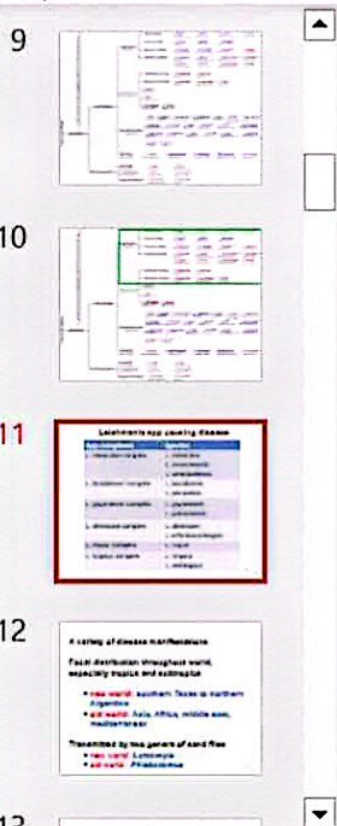
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Leishmania spp causing disease

Spp complexes	Species
L. <u>mexicana</u> complex	L. <u>mexicana</u> L. <u>amazonensis</u> L. <u>venezuelensis</u>
L. <u>braziliensis</u> complex	L. <u>braziliensis</u> L. <u>peruviana</u>
L. <u>guyanensis</u> complex	L. <u>guyanensis</u> L. <u>panamensis</u>
L. <u>donovani</u> complex	L. <u>donovani</u> L. <u>infantum/chagasi</u>
L. <u>major</u> complex	L. <u>major</u>
L. <u>tropica</u> complex	L. <u>tropica</u> L. <u>aethiopica</u>

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A variety of disease manifestations

Focal distribution throughout world, especially tropics and subtropics

= **new world**: southern Texas to northern Argentina

= **old world**: Asia, Africa, middle east, mediterranean

Transmitted by two genera of sand flies

= **new world**: *Lutzomyia*

= **old world**: *Phlebotomus*

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= **new world**: *Lutzomyia*

= **old world**: *Phlebotomus*

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Leishmania spp. causing disease

Leishmania spp.	Geographic distribution	Clinical features
L. tropica	Mediterranean, Middle East, Central Asia	Cutaneous leishmaniasis (CL)
L. major	Mediterranean, Middle East, Central Asia	Cutaneous leishmaniasis (CL)
L. infantum	Mediterranean, Middle East, Central Asia	Cutaneous leishmaniasis (CL), Visceral leishmaniasis (VL)
L. donovani	India, Bangladesh, Nepal, Pakistan	Visceral leishmaniasis (VL)
L. braziliensis	South America	Cutaneous leishmaniasis (CL), Mucocutaneous leishmaniasis (MCL), Visceral leishmaniasis (VL)
L. guyanaensis	South America	Cutaneous leishmaniasis (CL)
L. mexicana	Central America, Mexico	Cutaneous leishmaniasis (CL), Mucocutaneous leishmaniasis (MCL)
L. panamensis	Central America, Mexico	Cutaneous leishmaniasis (CL)
L. mexicana	Central America, Mexico	Cutaneous leishmaniasis (CL)
L. mexicana	Central America, Mexico	Cutaneous leishmaniasis (CL)

A variety of disease manifestations
 Total distribution throughout world, especially tropics and subtropics

- **new world:** southern Texas to southern Argentina
- **old world:** Asia, Africa, middle east, Mediterranean

Transmitted by two genera of sand flies

- **new world:** Lutzomyia
- **old world:** Phlebotomus

Clinical Spectrum of Leishmaniasis

Cutaneous Leishmaniasis (CL)

- **new world:** usually benign, self-healing skin lesions (aka, localized or simple CL)
- **old world:** usually benign, self-healing skin lesions (aka, localized or simple CL)

Diffuse Cutaneous Leishmaniasis (DCL)

- **new world:** usually benign, self-healing skin lesions (aka, localized or simple CL)
- **old world:** usually benign, self-healing skin lesions (aka, localized or simple CL)

Mucocutaneous Leishmaniasis (MCL)

- **new world:** usually benign, self-healing skin lesions (aka, localized or simple CL)
- **old world:** usually benign, self-healing skin lesions (aka, localized or simple CL)

Visceral Leishmaniasis (VL)

- **new world:** usually benign, self-healing skin lesions (aka, localized or simple CL)
- **old world:** usually benign, self-healing skin lesions (aka, localized or simple CL)

Clinical Spectrum of Leishmaniasis

Cutaneous Leishmaniasis (LCL)

= most common form, relatively benign self-healing skin lesions (aka, localized or simple CL)

Diffuse Cutaneous Leishmaniasis (DCL)

= rare cutaneous infection with non-ulcerating nodules resembling lepromatous leprosy

Mucocutaneous Leishmaniasis (MCL)

= simple skin lesions that metastasize to mucosae (especially nose and mouth region)

Visceral Leishmaniasis (VL), aka, kala-azar

= generalized infection of the RES, high mortality
 = PKDL (post-kala-azar dermal leishmanoids)

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Leishmania spp causing disease

Spp complexes	Species	Diseases
L. mexicana complex	L. mexicana L. amazonensis L. venezuelensis	LCL, DCL LCL, DCL I LCL
L. braziliensis complex	L. braziliensis L. peruviana	LCL, MCL LCL
L. guyanensis complex	L. guyanensis L. panamenensis	LCL LCL
L. donovani complex	L. donovani L. infantum/chagasi	VL, MCL VL, LCL, MCL
L. major complex	L. major	LCL
L. tropica complex	L. tropica L. aethiopica	LCL LCL, MCL, DCL

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37%

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Clinical Spectrum of Leishmaniasis

Volume

15



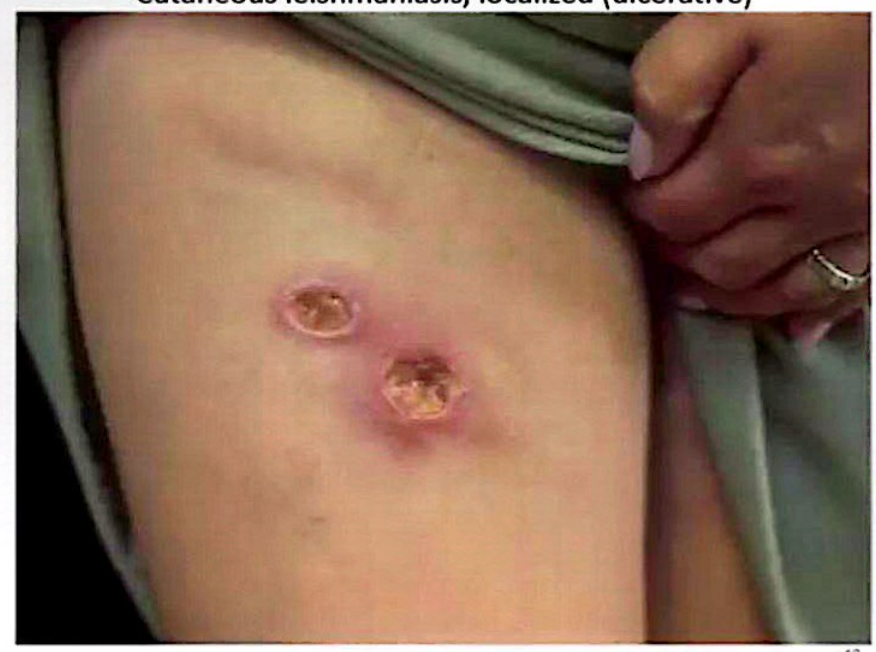
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Cutaneous leishmaniasis, localized (ulcerative)

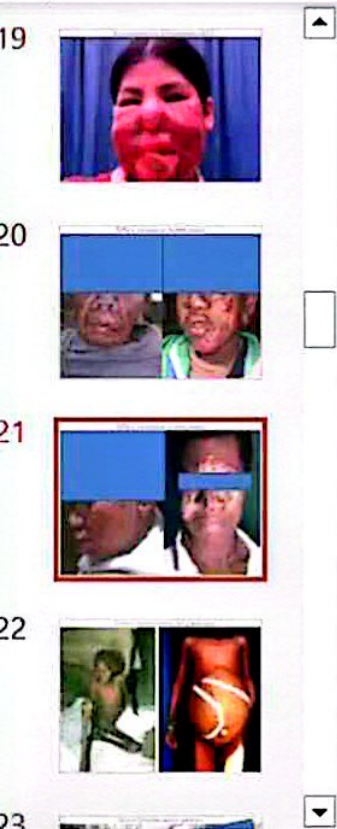


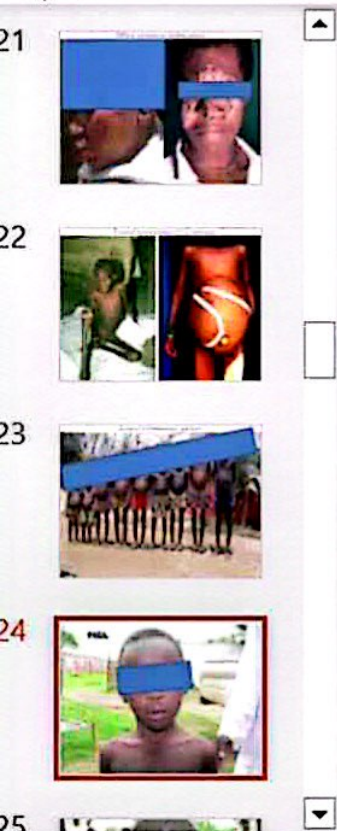
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Mucocutaneous leishmaniasis (MCL)

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Diagnosis of CL

Clinical dx: skin lesion:
chronic, painless, 'clean' ulcer
nasopharyngeal lesions
nodular lesions

Parasitology: smear & culture (NNN medium)
[dermal scrapping, biopsy]

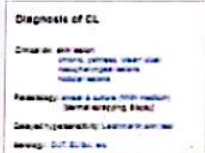
Delayed hypersensitivity: Leishmanin skin test

Serology: DAT, ELISA, etc.

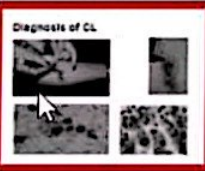
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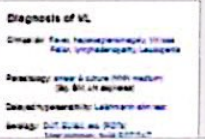
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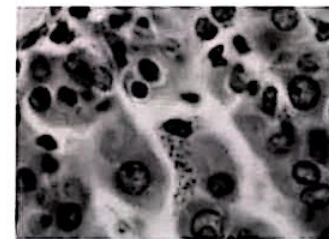
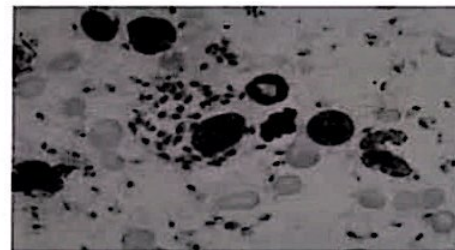
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Diagnosis of CL



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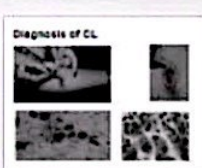
Visceral leishmaniasis

Characteristics:

- Untreated, fatal (97%)
- Conventional diagnosis, cumbersome
- Treatment toxic, cumbersome
- HIV co-infection common
- Epidemic potential

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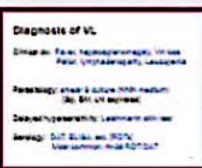
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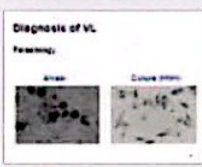
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Diagnosis of VL

Clinical dx: Fever, hepatosplenomegaly, Wt loss
Pallor, lymphadenopathy, Leucopenia

Parasitology: smear & culture (NNN medium)
[Sp, BM, LN aspirates]

Delayed hypersensitivity: Leishmanin skin test

Serology: DAT, ELISA, etc. [RDTs]
Most common: rk-39 RDT/DAT

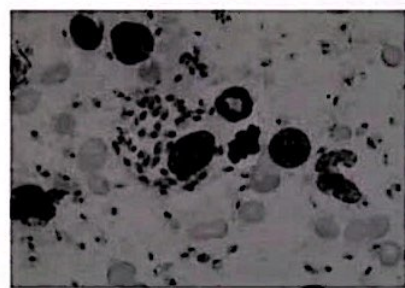
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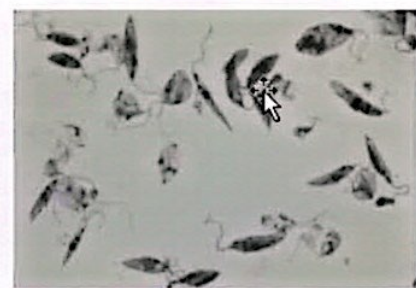
Diagnosis of VL

Parasitology

Smear



Culture (NNN)



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Diagnosis

Challenges

- Clinical diagnosis: Inaccurate
40 - 60% of clinical suspects
- Parasitological Invasive, painful, risky techniques:
- Serological tests: Variable intrinsic validity
- Skills: Inadequate

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Diagnosis of VL
 Smear: Taper technique, 100x
 Bone marrow aspirate: 100x
 Bone marrow aspirate: 100x
 Bone marrow aspirate: 100x
 Bone marrow aspirate: 100x

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Diagnosis of VL
 Bone marrow aspirate: 100x
 Bone marrow aspirate: 100x

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Diagnosis
 Bone marrow aspirate: 100x
 Bone marrow aspirate: 100x
 Bone marrow aspirate: 100x
 Bone marrow aspirate: 100x

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Parasitological diagnosis
 Comparison of source of tissue

Type of tissue	Recovery rate (%)
Spleen	96 - 98
BM	52 - 86 60 - 70
LN	52 - 78
Liver	70 - 90

Parasitological diagnosis

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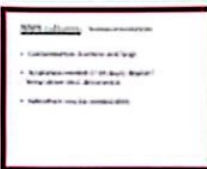
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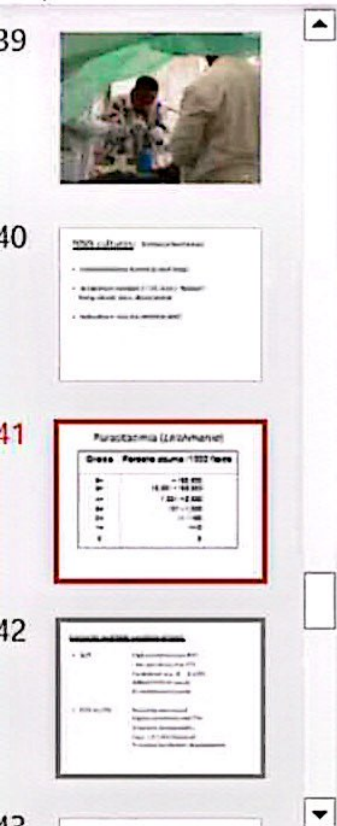
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NNN cultures: Technical limitations

- Contamination: bacteria and fungi
- Incubation needed (7-10 days): Repeat?
Temp above 28°C detrimental
- Subculture may be needed (BM)

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Parasitaemia (*Leishmania*)

Grade	Parasite counts /1000 fields
6+	> 100,000
5+	10,001 - 100,000
4+	1,001 - 10,000
3+	101 - 1,000
2+	11 - 100
1+	1-10
0	0

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Currently available serological tests

- **DAT:**
 - High sensitivity and NPV
 - Low specificity and PPV
 - Prohibitive cost (6 – 8 USD)
 - Adhoc/limited supply
 - Microtitration system
- **RDТ (rk-39):**
 - Simplicity and speed
 - Higher specificity and PPV
 - Potential marketability
 - Cost, 2.0 USD (Diamed)
 - Potential for further improvement

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Treatment regimens

First line = gSodium Stibogluconate
20 mg/kg b.wt, 30 days

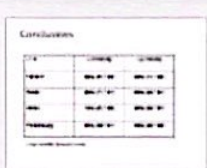
Second-line = Amphotericin B
1.0 mg/kg on alternate days (total 15 mg/kg)

Third-line = Miltefosine
50 mg/day for 28 days, < 25 kg
100 mg/day for 28 days, >25 kg
= Paromomycin, combination with SSG
PM 15 mg/kg + gSSG, 17 days
= Liposomal Amphotericin B (AmB)
3-5 mg/kg, 6 days (days 1,2,3,5,10,15)

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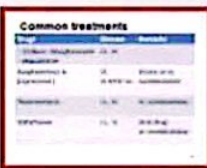
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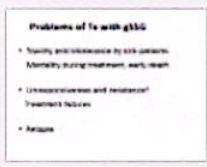
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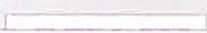
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Common treatments

Drugs	Disease	Remarks
= Sodium stibogluconate = <u>Glucantime</u>	CL, VL	
<u>Amphotericin B</u> (Liposomal)	VL VL+HIV co	Mono or in combinations
<u>Paromomycin</u>	CL, VL	In combinations
<u>Miltefosine</u>	CL, VL	Oral drug In combinations

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Problems of Tx with gSSG

- Toxicity and intolerance by sick patients
Mortality during treatment, early death
- Unresponsiveness and resistance?
Treatment failures
- Relapse

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Common treatments		
Drug	Dose	Frequency
Leishmanin	10 mg/kg	10 mg/kg
Leishmanin B	10 mg/kg	10 mg/kg
Leishmanin C	10 mg/kg	10 mg/kg
Leishmanin D	10 mg/kg	10 mg/kg
Leishmanin E	10 mg/kg	10 mg/kg
Leishmanin F	10 mg/kg	10 mg/kg
Leishmanin G	10 mg/kg	10 mg/kg
Leishmanin H	10 mg/kg	10 mg/kg
Leishmanin I	10 mg/kg	10 mg/kg
Leishmanin J	10 mg/kg	10 mg/kg
Leishmanin K	10 mg/kg	10 mg/kg
Leishmanin L	10 mg/kg	10 mg/kg
Leishmanin M	10 mg/kg	10 mg/kg
Leishmanin N	10 mg/kg	10 mg/kg
Leishmanin O	10 mg/kg	10 mg/kg
Leishmanin P	10 mg/kg	10 mg/kg
Leishmanin Q	10 mg/kg	10 mg/kg
Leishmanin R	10 mg/kg	10 mg/kg
Leishmanin S	10 mg/kg	10 mg/kg
Leishmanin T	10 mg/kg	10 mg/kg
Leishmanin U	10 mg/kg	10 mg/kg
Leishmanin V	10 mg/kg	10 mg/kg
Leishmanin W	10 mg/kg	10 mg/kg
Leishmanin X	10 mg/kg	10 mg/kg
Leishmanin Y	10 mg/kg	10 mg/kg
Leishmanin Z	10 mg/kg	10 mg/kg

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Problems of Tx with SSG	
• Toxicity and intolerance by sick patients	• Mortality during treatment, early death
• Unresponsiveness and resistance?	• Treatment failure
• Relapse	

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SSG toxicity	
• Cardiac	
• Renal	
• Hepatic	
• Pancreatic	

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Intolerance to SSG	
• Age	
• Severity of illness	
• Concomitant infections	
• IxV co-infection	

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SSG toxicity

- Cardiac
- Renal
- Hepatic
- Pancreatic

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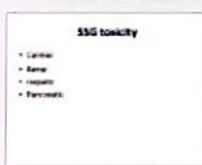
Intolerance to SSG

Risk factors

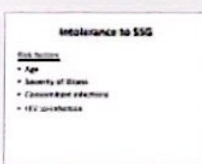
- Age
- Severity of illness
- Concomitant infections
- HIV co-infection

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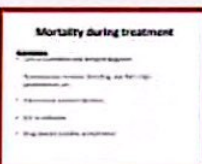
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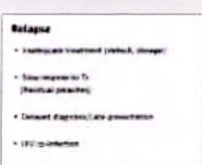
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Mortality during treatment

Risk factors

- Late presentation and delayed diagnosis

Pancytopenia, Anemia, Bleeding, low BMI, high parasitaemia, etc..

- Intercurrent events/infections
- HIV co-infection
- Drug toxicity (cardiac arrhythmias)

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Relapse

- Inadequate treatment (default, dosage)
- Slow response to Tx
[Residual parasites]
- Delayed diagnosis/Late presentation
- HIV co-infection

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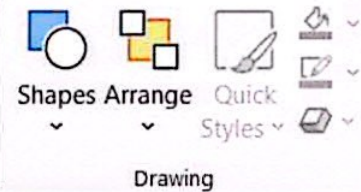
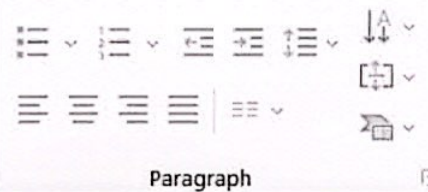
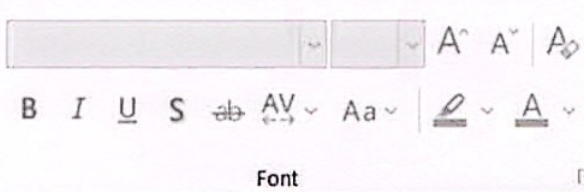
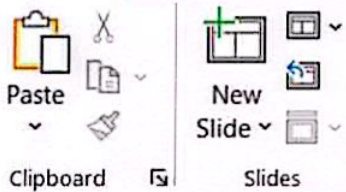
PKDL

Aetiology

- Strain versus host
- Environment
- SSG versus other drugs??
- HIV co-infection

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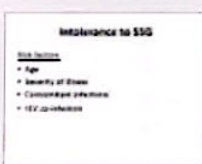
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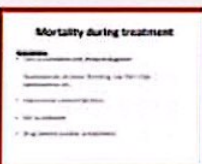
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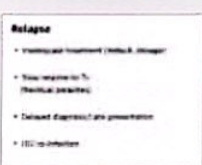
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Mortality during treatment

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Pancytopenia, Anemia, Bleeding, low BMI, high parasitaemia, etc..

- Intercurrent events/infections
- HIV co-infection
- Drug toxicity (cardiac arrhythmias)

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49 Mortality during treatment

50 Relapse

51 PKDL

52 HIV co-infection of VL

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PKDL

Aetiology

- Strain versus host
- Environment
- SSG versus other drugs??
- HIV co-infection

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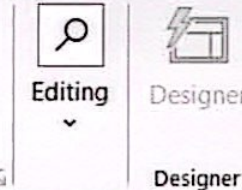
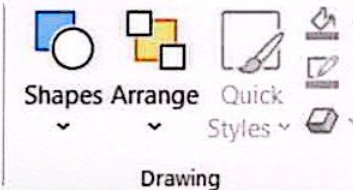
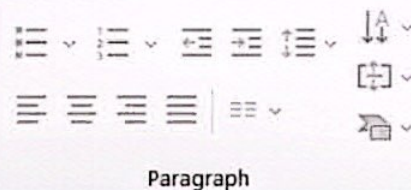
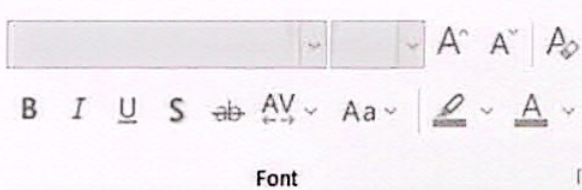
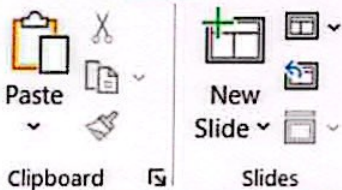
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Relapse

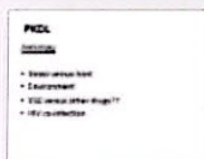
- Inadequate treatment (default, dosage)
- Slow response to Tx
[Residual parasites]
- Delayed diagnosis/Late presentation
- HIV co-infection

50

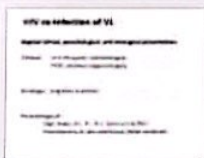
Click to add notes



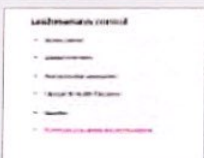
51



52



53



54



Enhanced case detection and treatment

Save lives, reduce mortality

Prevent accumulation of cases

Prevent lateral spread of infection

Reduce human reservoir

Mapping distribution and risk areas

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Click to add notes